

在线实训平台 环境操作展示文档

基于 GitLab + Jenkins + Docker Registry + Kubernetes 构建在线实训平台发布体系，实现代码托管、镜像构建、私有镜像仓库推送、K8s 滚动更新、统一 HTTPS 入口、监控日志观测和常见故障排查闭环。

1. 项目环境总览

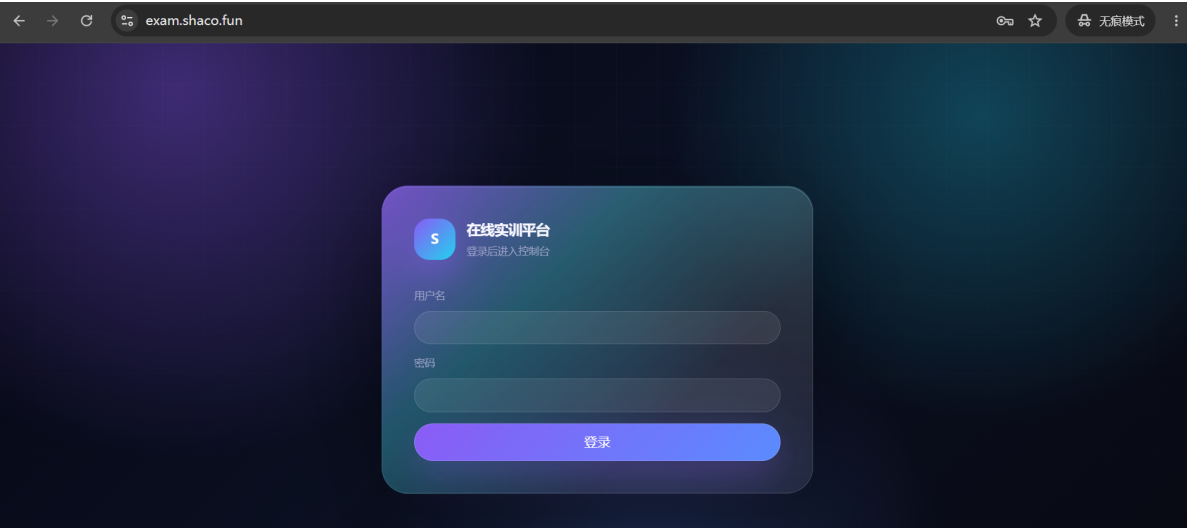
1.1 环境入口

系统	地址	说明
在线实训平台	<code>https://exam.shaco.fun</code>	平台业务访问入口
Jenkins	<code>https://jenkins.shaco.fun</code>	CI/CD 自动化构建与发布入口
Grafana	<code>https://grafana.shaco.fun</code>	监控看板与日志查询入口
GitLab	<code>https://git.shaco.fun</code>	代码仓库与镜像仓库入口

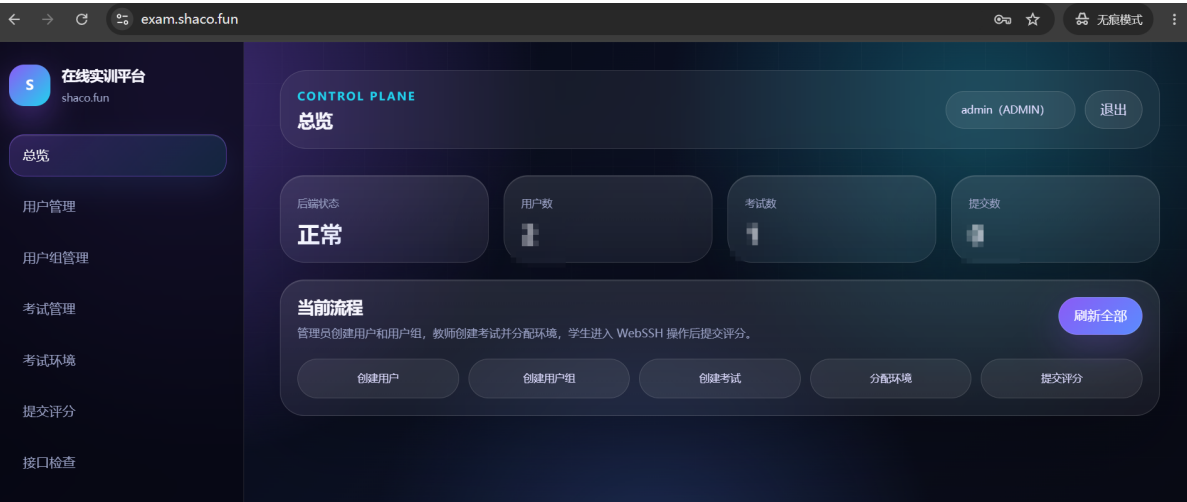
1.2 环境展示

本项目通过 `shacofun59` 作为统一 HTTPS 入口，对在线实训平台、Jenkins、Grafana、GitLab 等服务进行统一反向代理。业务访问、代码托管、自动化发布和监控观测均通过域名访问，降低了内部服务直接暴露风险，也方便后续维护和交付展示。

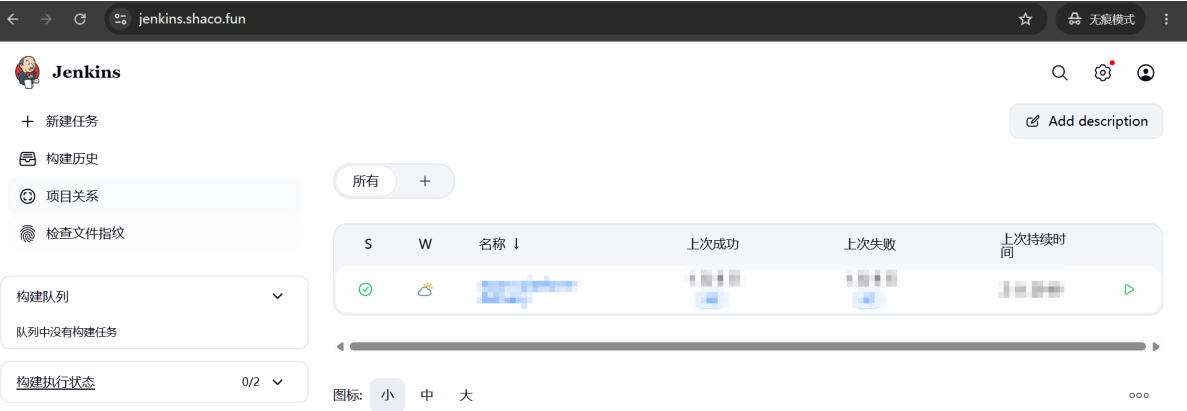
业务系统入口访问



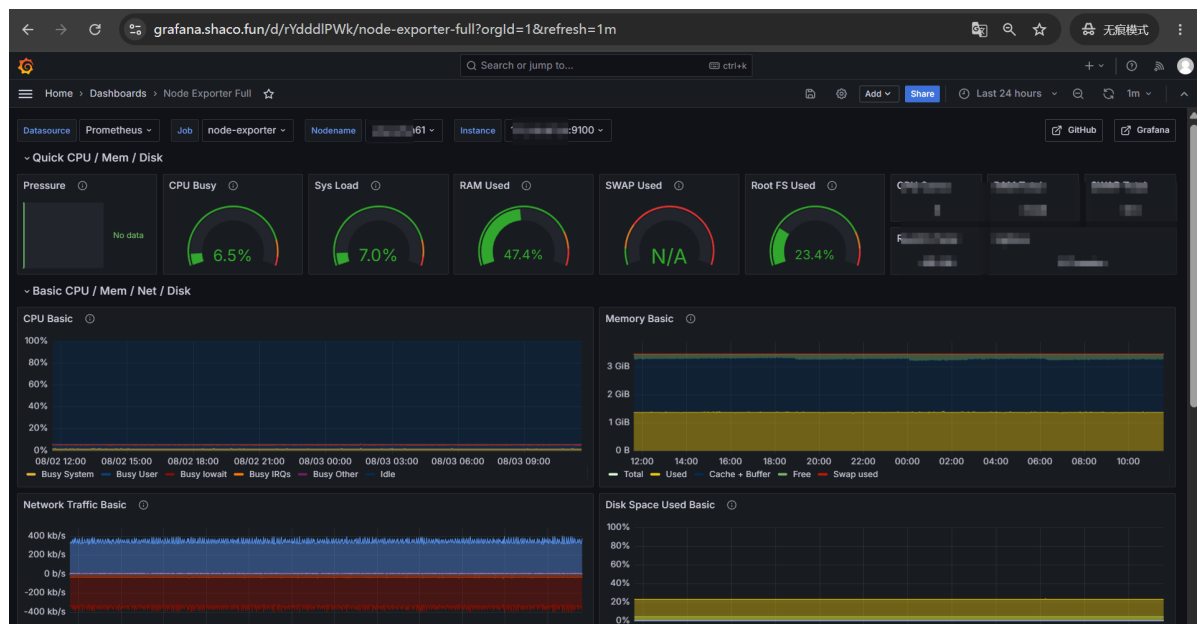
业务系统首页



自动化发布平台



监控平台



代码仓库

The screenshot shows the 'Your work / Projects' page in GitLab. The page displays a list of projects under the 'Contributed' tab. The projects are listed with their names, owners, and creation dates. The page also includes a search bar and filters for 'Starred', 'Personal', 'Member', and 'Inactive' projects. A notification banner at the top states: 'To help improve GitLab, we would like to periodically collect usage information. This can be changed at any time in your settings.' The page also includes a 'Send service data' button and a 'Don't send service data' button.

The screenshot shows the 'Container registry' page in GitLab. The page displays a list of image repositories under the 'Container registry' tab. The repositories are listed with their names, owners, and creation dates. The page also includes a search bar and filters for 'Updated' and 'Filter results'. A notification banner at the top states: 'The next-generation container registry is now available for upgrade on self-managed instances. This upgraded registry supports online garbage collection, and has significant performance and reliability improvements.' The page also includes a 'CLI Commands' button and a 'Set up cleanup' button.

2. 架构说明

2.1 架构组成

本项目整体采用：

GitLab

-> Jenkins Pipeline

-> Docker 镜像构建

-> GitLab Container Registry

-> Kubernetes 滚动发布

-> Prometheus/Grafana/Loki 监控日志观测

2.2 展示

```
[root@shacofun61 ~]# kubectl get nodes -o wide
```

NAME	STATUS	ROLES	AGE	VERSION	INTERNAL-IP	EXTERNAL-IP	OS-IMAGE	KERNEL-VERSION	CONTAINER-RUNTIME
shacofun61	Ready	control-plane	1	v1.3	10.1.1.61	<none>	Rocky Linux	4.18.0-147.el8.x86_64	cri-o
shacofun62	Ready	<none>	1	v1.3	10.1.1.62	<none>	Rocky Linux	4.18.0-147.el8.x86_64	cri-o
shacofun63	Ready	<none>	1	v1.3	10.1.1.63	<none>	Rocky Linux	4.18.0-147.el8.x86_64	cri-o
shacofun64	Ready	<none>	1	v1.3	10.1.1.64	<none>	Rocky Linux	4.18.0-147.el8.x86_64	cri-o
shacofun65	Ready	<none>	1	v1.3	10.1.1.65	<none>	Rocky Linux	4.18.0-147.el8.x86_64	cri-o

```
[root@shacofun61 ~]# kubectl -n exam-platform get deploy.svc.pod -o wide
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES	SELECTOR
deployment.apps/backend	1/1	1	1	1m	backend	registry.shaco.fur	app=exam
deployment.apps/frontend	1/1	1	1	1m	frontend	registry.shaco.fur	app=exam
deployment.apps/redis	1/1	1	1	1m	redis	registry.shaco.fur	app=exam
deployment.apps/student-agent	1/1	1	1	1m	student-agent	registry.shaco.fur	app=exam

```
[root@shacofun61 ~]# kubectl -n exam-platform get svc.pod -o wide
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE	SELECTOR
service/backend	NodePort	10.1.1.61	<none>	8080/TCP	1m	app=exam
service/frontend	NodePort	10.1.1.62	<none>	8080/TCP	1m	app=exam
service/redis	ClusterIP	10.1.1.63	<none>	6379/TCP	1m	app=exam
service/student-agent	NodePort	10.1.1.64	<none>	8080/TCP	1m	app=exam
service/exam	ClusterIP	10.1.1.65	<none>	8080/TCP	1m	app=exam

```
[root@shacofun61 ~]# kubectl -n exam-platform get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	READINESS
pod/backend	1/1	Running	0	1m	10.1.1.61	shacofun61	<none>	<none>
pod/frontend	1/1	Running	0	1m	10.1.1.62	shacofun62	<none>	<none>
pod/redis	1/1	Running	0	1m	10.1.1.63	shacofun63	<none>	<none>
pod/student-agent	1/1	Running	0	1m	10.1.1.64	shacofun64	<none>	<none>
pod/exam	1/1	Running	0	1m	10.1.1.65	shacofun65	<none>	<none>

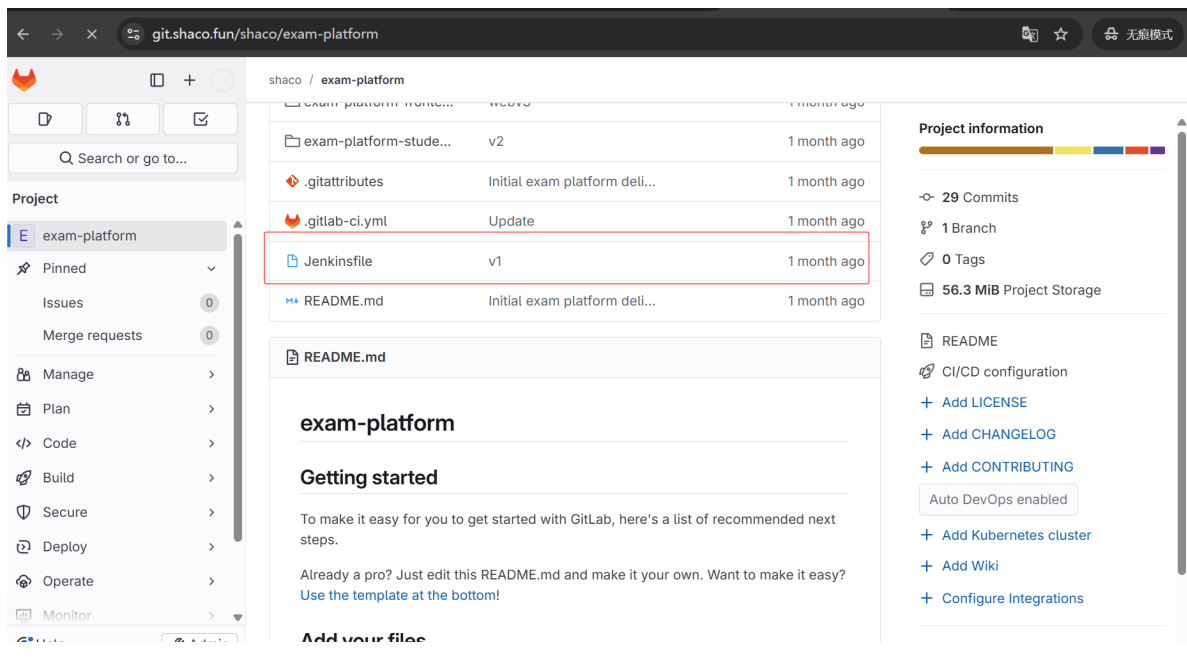
```
[root@shacofun61 ~]# kubectl -n kube-system get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	READINESS
calico-kube-controllers	1/1	Running	0	1m	10.1.1.61	shacofun61	<none>	<none>
calico-node	1/1	Running	0	1m	10.1.1.61	shacofun61	<none>	<none>
calico-node	1/1	Running	0	1m	10.1.1.62	shacofun62	<none>	<none>
calico-node	1/1	Running	0	1m	10.1.1.63	shacofun63	<none>	<none>
calico-node	1/1	Running	0	1m	10.1.1.64	shacofun64	<none>	<none>
calico-node	1/1	Running	0	1m	10.1.1.65	shacofun65	<none>	<none>
coredns	1/1	Running	0	1m	10.1.1.61	shacofun61	<none>	<none>
coredns	1/1	Running	0	1m	10.1.1.62	shacofun62	<none>	<none>
etcd	1/1	Running	1	1m	10.1.1.61	shacofun61	<none>	<none>
kube-apiserver	1/1	Running	1	1m	10.1.1.61	shacofun61	<none>	<none>
kube-controller-manager	1/1	Running	1	1m	10.1.1.61	shacofun61	<none>	<none>
kube-proxy	1/1	Running	0	1m	10.1.1.61	shacofun61	<none>	<none>
kube-proxy	1/1	Running	0	1m	10.1.1.62	shacofun62	<none>	<none>
kube-proxy	1/1	Running	0	1m	10.1.1.63	shacofun63	<none>	<none>
kube-proxy	1/1	Running	0	1m	10.1.1.64	shacofun64	<none>	<none>
kube-proxy	1/1	Running	0	1m	10.1.1.65	shacofun65	<none>	<none>
kube-scheduler	1/1	Running	1	1m	10.1.1.61	shacofun61	<none>	<none>

平台从早期 Docker Compose 部署逐步升级到 Kubernetes 多节点集群部署。Kubernetes 负责服务编排、Pod 调度、滚动更新和资源隔离；Jenkins 负责自动化构建发布；GitLab 和 Registry 负责代码与镜像管理；Prometheus、Grafana、Loki 负责指标和日志观测

3. GitLab 代码仓库使用

项目代码托管在 **GitLab** 中，前端、后端、执行面和学生镜像按目录拆分。
Jenkinsfile 位于仓库根目录，用于定义自动化构建、镜像推送和 **Kubernetes** 发布流程。通过 **GitLab + Registry** 实现代码、镜像和部署配置的统一管理。



4. Jenkins CI/CD 发布流程

4.1 Jenkins 入口

<https://jenkins.shaco.fun>

 Jenkins / exam-platform-delivery

状态

变更历史

Build with Parameters

配置

删除 Pipeline

Stages

重命名

流水线语法

 exam-platform-delivery

Pipeline script from SCM

相关链接

最近一次构建(#9), 1月9日之前

最近稳定构建(#9), 1月9日之前

最近成功的构建(#9), 1月9日之前

最近失败的构建(#7), 1月9日之前

最近未成功的构建(#7), 1月9日之前

最近完成的构建(#9), 1月9日之前

Builds >

Q 过滤构建...

/

2026年6月24日

 #9 下午5:43

 #8 下午5:40

4.2 Jenkins 发布后验证命令

在 Kubernetes 控制节点执行：

```
[root@shacofun61 ~]# kubectl -n exam-platform get deploy backend frontend student-agent \
-o custom-columns=NAME:.metadata.name,IMAGE:.spec.template.spec.containers[0].image
NAME          IMAGE
backend       registry.shaco.fun/shaco/exam-platform-frontend
frontend      registry.shaco.fun/shaco/exam-platform-backend
student-agent  registry.shaco.fun/shaco/exam-platform-agent
```

查看滚动发布状态：

```
[root@shacofun61 ~]# kubectl -n exam-platform rollout status deployment/frontend
deployment "frontend" successfully rolled out
[root@shacofun61 ~]# kubectl -n exam-platform rollout status deployment/backend
deployment "backend" successfully rolled out
[root@shacofun61 ~]# kubectl -n exam-platform rollout status deployment/student-agent
deployment "student-agent" successfully rolled out
[root@shacofun61 ~]#
```

查看 Pod：

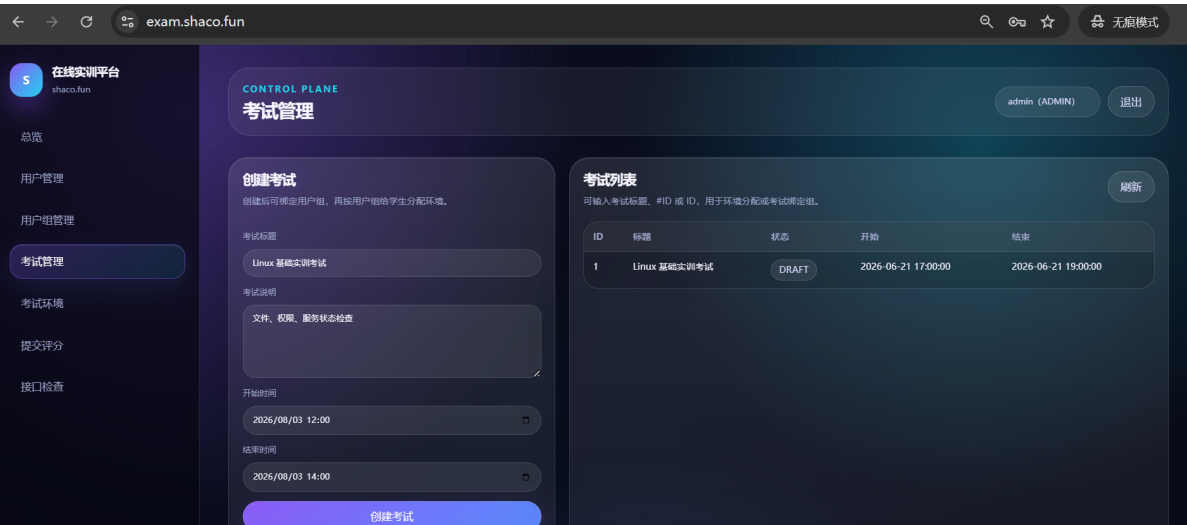
```
[root@shacofun61 ~]# kubectl -n exam-platform get pod -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE          NOMINATED NODE   READINESS GATES
bac-66666666  1/1     Running   0           1m    10.1.1.10     shaco-node-2  <none>           <none>
fro-66666666  1/1     Running   0           1m    10.1.1.11     shaco-node-2  <none>           <none>
red-66666666  1/1     Running   0           1m    10.1.1.12     shaco-node-2  <none>           <none>
stu-66666666  1/1     Running   0           1m    10.1.1.13     shaco-node-2  <none>           <none>
tty-66666666  1/1     Running   0           1m    10.1.1.14     shaco-node-3  <none>           <none>
tty-66666666  1/1     Running   0           1m    10.1.1.15     shaco-node-3  <none>           <none>
```

5. 在线实训平台业务入口展示

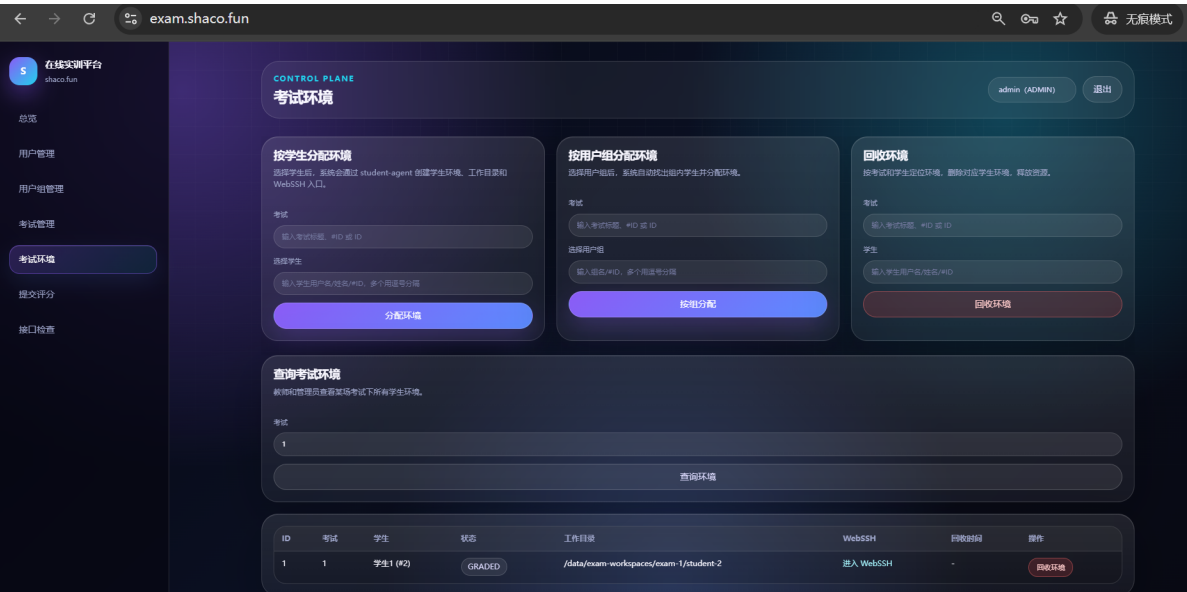
5.1 平台入口

https://exam.shaco.fun

考试/实训列表页面



创建或分配学生环境页面



WebSSH/ttyd 页面

